

InternVL3-1B5 Instruct

Fine-Tuning Detailed Report - SatQuery AI

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Table of Contents

	Section
1	Model Overview - InternVL3-1B5 Instruct
2	Why We Chose This Model
3	Datasets Used for Fine-tuning
4	Evaluation Metrics - Meaning and Formula
5	Fine-tuning Method - QLoRA Explained
6	Training Run V1 - First Attempt
7	Training Run V2 - Improved Attempt
8	Results - Before and After Fine-tuning
9	Comparison with EarthMind-4B
10	Conclusion and Final Decision

Section 1: Model Overview - InternVL3-1B5 Instruct

InternVL3-1B5 Instruct is a Vision-Language Model (VLM) developed by OpenGVLab at Shanghai AI Laboratory. It is part of the InternVL3 model family which represents the third generation of InternVL models. The "1B5" refers to 1.5 Billion parameters and "Instruct" means it has been instruction-tuned to follow natural language commands.

1.1 Model Links

Resource	Link
HuggingFace Model	huggingface.co/OpenGVLab/InternVL3-1B
Research Paper	arxiv.org/abs/2504.10479 - InternVL3: Exploring Advanced Training and Test-Time Recipes for Open-Source Multimodal Models
GitHub Repo	github.com/OpenGVLab/InternVL
Model Family	InternVL3 Series: 1B, 2B, 4B, 8B, 14B, 38B, 78B

1.2 Architecture

InternVL3 uses a dual-component architecture combining a Vision Encoder and a Language Model. The vision encoder is InternViT-300M which processes images into visual feature tokens. These tokens are then passed to a language model (Qwen2.5-1.5B) which generates text responses. The two components are connected via a Multi-Layer Perceptron (MLP) projection layer.

Component	Details	Parameters
Vision Encoder	InternViT-300M - processes satellite images into 256 visual tokens	300M
MLP Connector	Projects vision features to language model dimension space	~50M
Language Model	Qwen2.5-1.5B - generates natural language responses from visual + text tokens	1.5B
Total	Complete vision-language understanding and generation	~1.8B

1.3 Key Capabilities

Before fine-tuning, InternVL3-1B5 Instruct supports visual question answering, image captioning, object detection, document understanding, and multi-turn conversation with images. It was pre-trained on general internet images (natural photos, documents, charts) and NOT on satellite imagery specifically. This is why fine-tuning was needed for our satellite image analysis tasks.

Section 2: Why We Chose InternVL3-1B5 Instruct

The choice of InternVL3-1B5 Instruct was driven by the BigEarthNet.txt research paper (arxiv 2506.01667) which specifically used InternVL3-1B as the base model for satellite image fine-tuning. The paper demonstrated that fine-tuning InternVL3-1B on satellite data (called RS-InternVL after fine-tuning) achieved over 31% average improvement across all tasks. This paper-validated approach gave us confidence that this model family is suitable for remote sensing adaptation.

Reason	Explanation
Paper Validated	BigEarthNet.txt paper (2026) used InternVL3-1B and proved 31%+ improvement after satellite fine-tuning. This is the same dataset family we use for evaluation.
Right Size	1.5B parameters fits on RTX 4060 8GB VRAM with 4-bit quantization. Larger models like 7B+ would not fit.
Instruct Tuned	The Instruct variant follows natural language commands well, making it suitable for query-based satellite analysis.
Open Source	Fully open source on HuggingFace, free to download, fine-tune, and deploy without API costs.
Fast Inference	1.5B parameters means faster inference than EarthMind-4B, suitable for real-time analysis in the SatQuery AI pipeline.
LoRA Compatible	Architecture supports PEFT/LoRA fine-tuning enabling efficient training on RTX 4060 without full parameter updates.

2.1 BigEarthNet.txt Paper Results for InternVL3

The paper that motivated our choice showed the following improvements when fine-tuning InternVL3-1B on BigEarthNet.txt satellite data. These numbers represent what was achievable with proper fine-tuning and served as our target benchmark.

Task	Before FT	After FT	Improvement
Binary VQA	61.96%	73.29%	+11.3%
MCQ	37.55%	51.49%	+13.9%
Captioning	0.96%	34.04%	+3300%
Ref Detection	31.73%	65.84%	+107%

Section 3: Datasets Used for Fine-tuning

Four datasets were used to create the unified training data for InternVL3 fine-tuning. Each dataset covers different satellite image analysis tasks. The datasets were combined into a unified training manifest of approximately 30,000 multi-modal samples.

3.1 Dataset Summary

Dataset	HuggingFace Link	Task	Samples	Images?
VRSBench	xiang709/VRSBench	VQA + Captioning + Grounding	29,614	Yes - PNG
SECOND	EVER-Z/torchance_second	Change VQA (bi-temporal)	2,968 pairs	Yes - PNG
BigEarthNet.txt	BIFOLD-BigEarthNetv2-0/BigEarthNet.txt	VQA + Captioning (text only)	9,553,962	No - Text only
RSVQA-LR-2k	dmarsili/RSVQA-LR-2k	Count and Area VQA	2,000	Yes - embedded

3.2 Dataset Details

VRSBench

VRSBench is the primary evaluation and training dataset. It contains 29,614 satellite images with corresponding question-answer pairs, captions, and region grounding annotations. The train split was used for fine-tuning while the test split was used for evaluation. It covers diverse satellite scenes including urban areas, agricultural land, water bodies, forests, and transportation infrastructure.

SECOND Dataset

SECOND (Self-supervised Semantic Change Detection) provides bi-temporal satellite image pairs showing the same geographic location at two different times. Each pair includes a before image, an after image, and a pixel-level change mask showing what type of land cover changed. We used this dataset to train InternVL3 for change detection visual question answering. The change mask was used to derive ground truth answers (yes = change detected, no = no change).

BigEarthNet.txt

BigEarthNet.txt contains 9.5 million text-only Q&A pairs derived from Sentinel-1 and Sentinel-2 satellite data. Importantly, this dataset does NOT include actual images in the downloaded version - it contains only the text annotations. We used it as text-only augmentation data to teach the model satellite domain vocabulary and land-use/land-cover (LULC) concepts without visual hallucination risks.

RSVQA-LR-2k

RSVQA Low-Resolution 2k dataset provides 2,000 samples of satellite image VQA focusing specifically on Count questions (how many objects) and Area questions (what area does X cover in square meters). These question types were identified as weak areas in the baseline evaluation where InternVL3 scored only 0% on area questions and 22.6% on count questions.

Section 4: Evaluation Metrics - Meaning and Formula

The following metrics were used to evaluate model performance before and after fine-tuning. Each metric captures a different aspect of model quality.

4.1 Accuracy Metrics (VQA Tasks)

Metric	Formula	What it Measures
Exact Match	Correct predictions / Total x 100	Prediction must exactly equal ground truth after normalization (lowercase, no punctuation).
Soft Accuracy	GT tokens in prediction / GT tokens	Checks if ground truth words appear anywhere in the prediction. Handles verbose model answers.
Numeric Accuracy	Matching numbers / Total	Extracts numbers from both prediction and GT. Used for count and area questions (e.g. 3, 495m2).
SBERT Cosine	<code>cosine_sim(embed(pred), embed(gt))</code>	Semantic similarity using Sentence-BERT 384-d vectors. Captures meaning even with different words.

4.2 Captioning Metrics

Metric	Formula	What it Measures
BLEU-4	$BP \times \exp(0.25 \times \text{sum of log n-gram precisions})$	Counts matching 1,2,3,4-gram sequences between prediction and reference caption. Strict metric.
ROUGE-L	LCS length / reference length	Longest Common Subsequence between prediction and reference. Measures sentence structure similarity.
METEOR	F-score with exact + stem + synonym match	Handles synonyms and word stemming. "urban" matches "city". More forgiving than BLEU.
CIDEr	TF-IDF weighted n-gram consensus score	Weights rare domain words higher. Satellite terms like wetlands and arable get more credit.
SBERT Cosine	<code>cosine_sim(embed(caption), embed(reference))</code>	Semantic similarity of entire caption. High score means model described same content differently.

4.3 Grounding Metrics

Metric	Formula	What it Measures
mIoU	Mean of (Intersection Area / Union Area) across all samples	How well predicted bounding box overlaps with ground truth box. 0=no overlap, 1=perfect.
Accuracy@50	% samples where IoU >= 0.50	Standard threshold. At least 50% overlap between predicted and ground truth box required.
Parse Success	% samples where valid bbox was extracted	Did the model output a valid [x1,y1,x2,y2] format? 100% means always outputs correct format.

Section 5: Fine-tuning Method - QLoRA Explained

We used QLoRA (Quantized Low-Rank Adaptation) to fine-tune InternVL3-1B5 on our satellite datasets. QLoRA combines two techniques: 4-bit quantization to reduce memory usage and LoRA to efficiently update only a small fraction of model parameters.

5.1 What is LoRA?

LoRA (Low-Rank Adaptation) adds small trainable matrices to existing model layers instead of updating all 1.5 billion parameters. Instead of updating weight matrix W directly, LoRA adds two small matrices A and B where the update is $A \times B$. The rank (r) determines how many parameters these matrices have. With rank=4, only about 3 million parameters are trained instead of 1.5 billion.

5.2 What is 4-bit Quantization?

Normal model weights use 32-bit or 16-bit floating point numbers. 4-bit quantization reduces each weight to just 4 bits, reducing memory usage by 4x to 8x. InternVL3-1B5 at full 16-bit precision needs about 3GB VRAM. With 4-bit quantization it fits in under 2GB VRAM, leaving plenty of room for activations during training on the RTX 4060 8GB.

5.3 V1 vs V2 Configuration Comparison

Setting	V1 (First Attempt)	V2 (Improved)	Why Changed
LoRA Rank	8	4	Lower rank = less overfitting risk
LoRA Alpha	32	16	Proportional to rank change
LoRA Dropout	0.1	0.3	Higher dropout prevents memorization
Learning Rate	2e-5	1e-5	Slower learning for better generalization
Warmup Steps	100	200	More warmup for stable training start
Gradient Clip	1.0	0.5	Tighter clipping prevents gradient explosion
Early Stopping	No	Yes	Stop before overfitting occurs
Validation Split	No	Yes - 10%	Monitor generalization during training
Replay Buffer	No	Yes - 20%	Prevent catastrophic forgetting
Trainable Params	~5.8M	~3M	Fewer params = less overfitting
Total Steps	1248	1404	More data but stopped early if needed

Section 6: Training Run V1 - First Attempt

The first fine-tuning attempt used standard LoRA settings without validation monitoring or early stopping. The training completed all 1248 steps but resulted in catastrophic overfitting where the model memorized training data and forgot its baseline capabilities.

6.1 V1 Loss Curve

Step	Training Loss	Observation
1	11.56	High initial loss - model has not learned anything yet
80	0.1507	Rapid improvement - model learning satellite patterns
500	0.2788	Loss increased slightly - possible hard batch
1000	0.0610	Good progress but getting low
1248	0.0000	DANGER - perfect zero loss means complete overfitting

Loss reaching 0.0000 is a warning sign in machine learning. It means the model has memorized every training sample perfectly but has lost the ability to generalize to new unseen images. This is called catastrophic overfitting.

6.2 What is Catastrophic Forgetting?

Catastrophic forgetting occurs when a neural network forgets previously learned information while learning new information. In our case, InternVL3 had learned general visual question answering from its pre-training. When we fine-tuned it on satellite data without safeguards, it overwrote those general capabilities with satellite-specific patterns. The result was that performance on the exact same VQA task actually became worse after fine-tuning (50% down to 28%).

Section 7: Training Run V2 - Improved Attempt

The second attempt incorporated all fixes identified from V1 - lower rank, higher dropout, slower learning rate, validation monitoring, early stopping, and a replay buffer to prevent catastrophic forgetting. Training ran for 7 hours 30 minutes.

7.1 V2 Complete Loss Curve

Step	Train Loss	Val Loss	Action
100	8.3228	8.1028	Training started - high loss normal
200	2.3358	1.4431	Best model saved - big improvement
300	0.4665	0.4323	Best model saved - healthy convergence
400	0.1179	0.1080	Best model saved - still improving
500	0.0097	0.0494	Best model saved - val still improving
600	0.0582	0.0385	Best model saved - good generalization
700	0.0299	0.0334	Best model saved - stable
800	0.0008	0.0312	Best model saved
900	0.0015	0.0293	Best model saved - lowest val loss so far
1000	0.0729	0.0298	Patience 1/3 - val did not improve
1100	0.1111	0.0297	Patience 2/3 - val barely changed
1200	0.0011	0.0287	Best model saved - new low!
1300	0.0014	0.0289	Patience 1/3 - training steps running out
1400	0.0007	0.0289	Patience 2/3 - all steps completed
1404	0.0001	0.0287	Training complete - best checkpoint: Step 1200

Key insight: In V2, validation loss stayed above 0.02 throughout training, never reaching zero. This means the model did not overfit completely. The best checkpoint was saved at Step 1200 with Val Loss = 0.0287.

Section 8: Results - Before and After Fine-tuning

The following tables show complete benchmark results for InternVL3-1B5 Instruct at three stages: Baseline (no fine-tuning), V1 Fine-tuned (overfit), and V2 Fine-tuned (improved). All benchmarks used 50 samples per task.

8.1 VQA Results

Metric	Baseline	V1 FT	V2 FT	Best
VRS VQA Soft	50.0%	28.0%	29.0%	Baseline
VRS VQA Exact	31.0%	27.0%	27.5%	Baseline
VRS SBERT	0.564	0.463	0.489	Baseline
RSVQA Soft	36.0%	32.8%	33.2%	Baseline
RSVQA Exact	22.6%	32.4%	33.2%	V2 FT
RSVQA Numeric	1.0%	0.0%	0.0%	Baseline
RSVQA SBERT	0.477	0.616	0.639	V2 FT

8.2 Captioning Results

Metric	Baseline	V1 FT	V2 FT	Best
BLEU-4	0.047	0.033	0.029	Baseline
ROUGE-L	0.247	0.224	0.220	Baseline
METEOR	0.264	0.268	0.268	V1/V2 FT
CIDEr	0.159	0.079	0.064	Baseline
SBERT	0.666	0.643	0.635	Baseline

8.3 Grounding and Change VQA Results

Metric	Baseline	V1 FT	V2 FT	Best
Grounding mIoU	0.140	0.082	0.083	Baseline
Grounding Acc@50	7.0%	3.5%	4.0%	Baseline
Change VQA Soft	99.5%	100.0%	100.0%	V1/V2 FT

Section 9: Comparison with EarthMind-4B

EarthMind-4B is our primary model which was pre-trained specifically on satellite imagery (Sentinel-1 SAR + Sentinel-2 optical) before we even started. Comparing InternVL3 fine-tuned results against EarthMind baseline shows why EarthMind was selected as the primary model for SatQuery AI.

Metric	EarthMind-4B	InternVL3 Base	InternVL3 V2 FT	Winner
VRS VQA Soft	72.0%	50.0%	29.0%	EarthMind
VRS VQA Exact	70.0%	31.0%	27.5%	EarthMind
VRS SBERT	0.880	0.564	0.489	EarthMind
RSVQA Soft	48.8%	36.0%	33.2%	EarthMind
RSVQA Exact	48.8%	22.6%	33.2%	EarthMind
RSVQA SBERT	0.719	0.477	0.639	EarthMind
Caption BLEU-4	0.085	0.047	0.029	EarthMind
Caption ROUGE-L	0.289	0.247	0.220	EarthMind
Caption METEOR	0.312	0.264	0.268	EarthMind
Caption CIDEr	0.259	0.159	0.064	EarthMind
Caption SBERT	0.732	0.666	0.635	EarthMind
Ground mIoU	0.324	0.140	0.083	EarthMind
Ground Acc@50	41.0%	7.0%	4.0%	EarthMind
Change VQA Soft	99.5%	99.5%	100.0%	InternVL3 FT
Recommendation	GO	PARTIAL	NO-GO	EarthMind

Section 10: Conclusion and Final Decision

10.1 Key Findings

#	Finding
1	Catastrophic forgetting occurred in both V1 and V2 fine-tuning attempts. InternVL3-1B5 at 1.5B parameters does not have enough capacity to learn satellite domain knowledge without overwriting its baseline VQA capabilities.
2	V2 improvements over V1 were marginal. Better hyperparameters (lower rank, higher dropout, early stopping) helped slightly but did not solve the fundamental forgetting problem.
3	Change VQA is the one task where fine-tuning helped. Both V1 and V2 achieved 100% soft accuracy on change detection, compared to 99.5% for EarthMind.
4	SBERT scores for RSVQA improved after fine-tuning (0.477 to 0.639), suggesting the model is semantically closer to correct answers even if exact match did not improve.
5	EarthMind-4B outperforms InternVL3 fine-tuned on 13 out of 15 metrics, despite EarthMind being evaluated without any task-specific fine-tuning.
6	The root cause is model architecture mismatch. EarthMind was designed and pre-trained specifically for satellite imagery (Sentinel-1 + Sentinel-2). InternVL3 was designed for general vision tasks.

10.2 Final Deployment Decision - Hybrid Architecture

Based on all fine-tuning experiments, the final deployment decision is a hybrid architecture where EarthMind-4B serves as the primary model for all tasks, while InternVL3-1B5 fine-tuned (V2) is used as a specialist for Change VQA where it achieved perfect performance. The LangGraph agentic controller routes queries to the appropriate model automatically.

Task	Model	Score	Reason
VQA	EarthMind-4B	72.0%	EarthMind wins by 43 percentage points over InternVL3 FT
Captioning	EarthMind-4B	BLEU 0.085	EarthMind produces better satellite domain captions
Grounding	EarthMind-4B	mIoU 0.324	EarthMind Acc@50 is 41% vs InternVL3 FT 4%
Change VQA	InternVL3 FT V2	100%	Fine-tuning improved this specific task. Lighter model is faster.
SAR + Optical	EarthMind-4B	N/A	EarthMind has built-in SAR support. InternVL3 does not.

10.3 Future Work

Option	Description	Expected Outcome
Larger Model	Try InternVL3-8B instead of 1B5. More parameters = less forgetting risk.	Better satellite VQA without catastrophic forgetting
DARE / TIES Merging	Model merging technique that combines pre-trained and fine-tuned weights mathematically.	Preserve baseline while adding satellite knowledge

More Diverse Data	Add DIOR-RSVG for grounding, RSICD for captioning, LEVIR-CD for change detection.	Better task-specific performance
EarthMind Fine-tune	Apply same QLoRA fine-tuning to EarthMind-4B which already has satellite knowledge.	Highest potential improvement - satellite base + task fine-tuning